

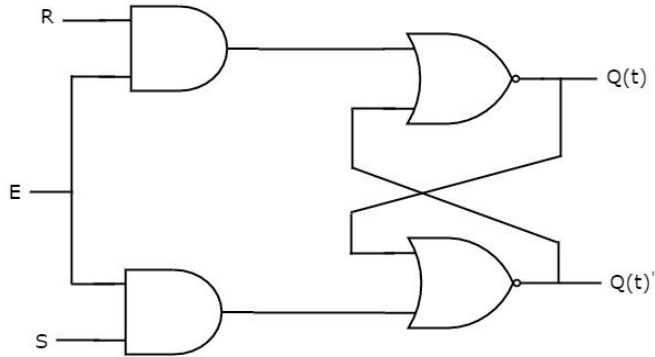


Flip-Flops & Sequential Circuits

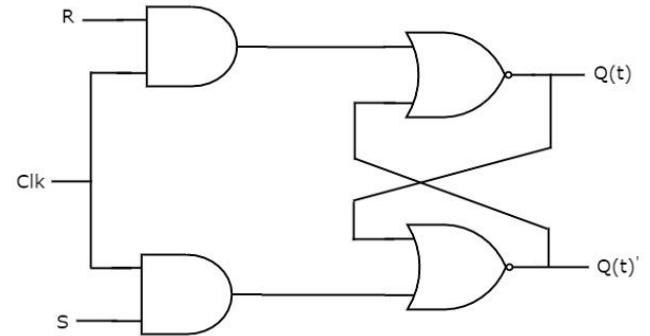
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Flip Flop vs Latches

Latches are level triggered

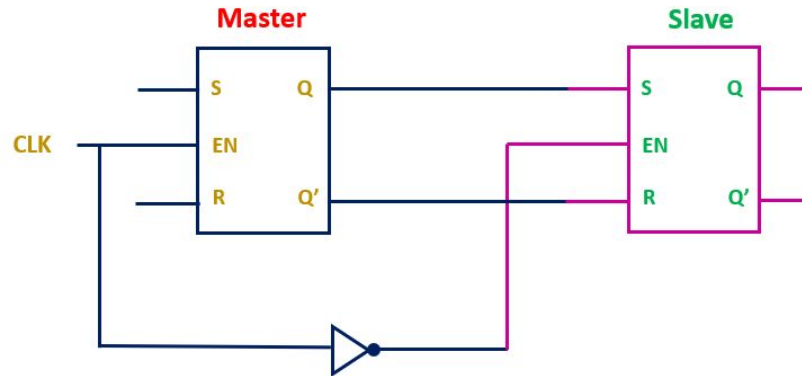


Flip Flops are Edge triggered

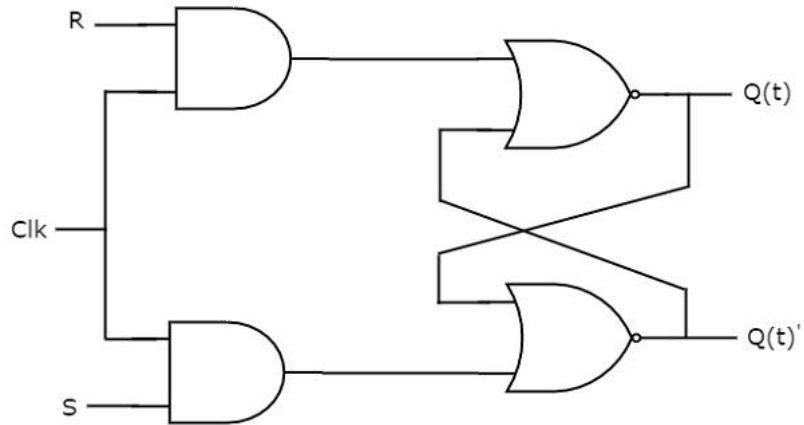


Flip Flop from Latches

- Flip Flops are made using latches placed in a Master-Slave configuration.
- This converts the state to change on only when the signal moves from low to high, as opposed to all the time when the enable signal is high



SR Flip Flop



S	R	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	Undefined

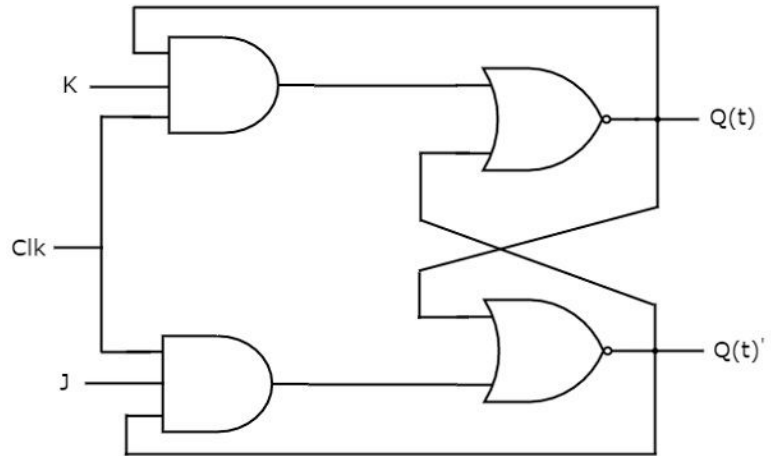


SR Flip Flop



```
1  module sr_ff (  
2      input s,  
3      input r,  
4      input clk,  
5      output reg q,  
6      output reg qbar  
7  );  
8      always @(posedge clk) begin  
9          case ({s,r})  
10             2'b00: {q,qbar} <= {q,qbar};  
11             2'b01: {q,qbar} <= 2'b01;  
12             2'b10: {q,qbar} <= 2'b10;  
13             2'b11: {q,qbar} <= 2'bxx;  
14          endcase  
15      end  
16  endmodule  
17
```

JK Flip Flop



J	K	Q(t+1)
0	0	Q(t)
0	1	0
1	0	1
1	1	Q(t)'



Excitation & Characteristic Tables

Q(t)	Q(t+1)	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

Q(t)	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

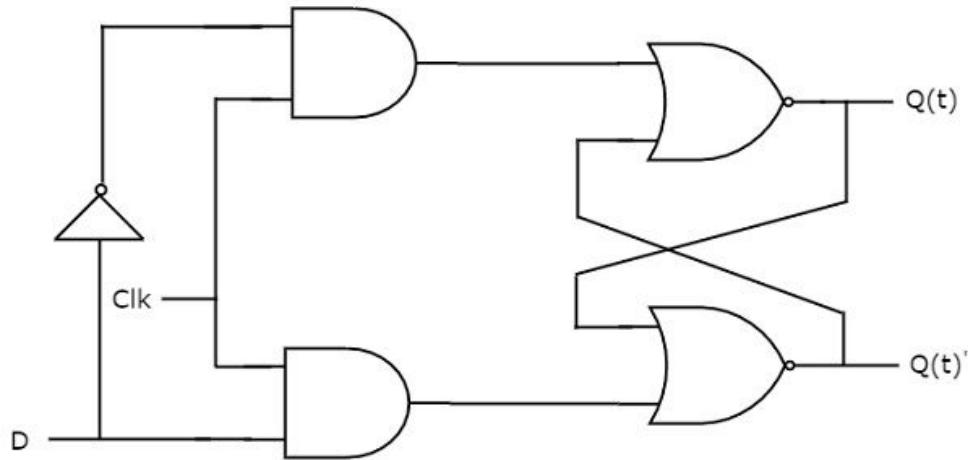


JK Flip Flop



```
1  module jk_ff (  
2      input j,  
3      input k,  
4      input clk,  
5      output reg q,  
6      output reg qbar  
7  );  
8      always @(posedge clk) begin  
9          case ({j,k})  
10             2'b00: {q,qbar} <= {q,qbar};  
11             2'b01: {q,qbar} <= 2'b01;  
12             2'b10: {q,qbar} <= 2'b10;  
13             2'b11: {q,qbar} <= ~{q,qbar};  
14          endcase  
15      end  
16  
17  endmodule
```

D Flip Flop



D	$Q(t+1)$
0	0
1	1



Excitation & Characteristic Tables

Q(t)	Q(t+1)	D
0	0	0
0	1	1
1	0	0
1	1	1

Q(t)	D	Q(t+1)
0	0	0
0	1	1
1	0	0
1	1	1

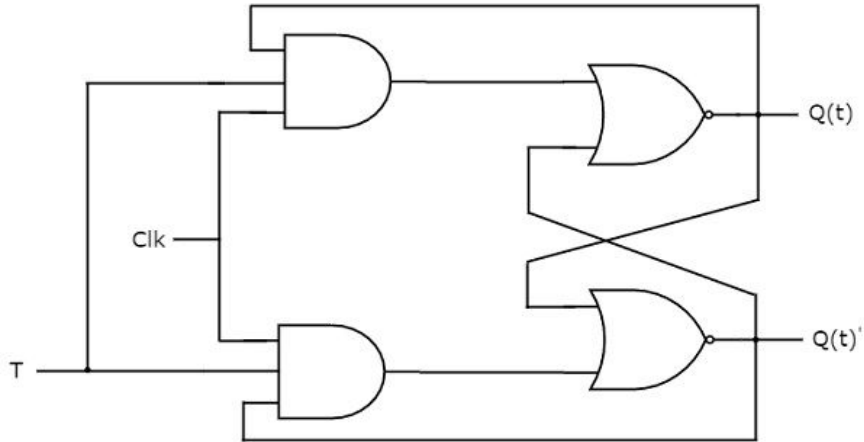


D Flip Flop



```
1  module d_ff (  
2      input d,  
3      input clk,  
4      output reg q,  
5  );  
6      always @(posedge clk) begin  
7          q <= d;  
8      end  
9  );  
10  
11 endmodule
```

T Flip Flop



T	Q(t+1)
0	Q(t)
1	Q(t)'



Excitation & Characteristic Tables

Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0

Q(t)	T	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0



T Flip Flop



```
1  module t_ff (  
2      input t,  
3      input clk,  
4      output reg q  
5  );  
6      always @(posedge clk) begin  
7          if (t)  
8              q <= ~q;  
9      end  
10 endmodule
```



Blocking vs Non-Blocking Assignment

- Blocking assignment ($=$) immediately evaluates the RHS and immediately assigns it into the LHS.
- Blocking assignment is best used while making combinatorial circuits involving continuous assignments.



Blocking vs Non-Blocking Assignment

- Non-blocking assignment ($\leq$) evaluates the RHS but does not update the LHS till all the RHS expressions are evaluated and then does the assignment.
- Non-blocking assignments are used to make sequential circuits.



Blocking vs Non-blocking

All a, b and c start with the value 5



```
1 a = 8'd5;    // blocking: updates a immediately
2 b = b + a;    // sees the new a (5) -> b = 10
3 c = c + b;    // sees the new b (10) -> c = 15
```



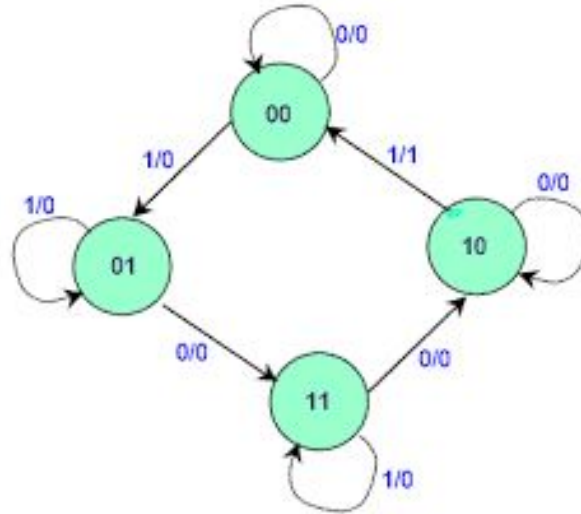
```
1 a <= 8'd5;    // non-blocking: schedule a = 5
2 b <= b + a;    // RHS uses OLD b and OLD a (both 5) -> b = 10
3 c <= c + b;    // RHS uses OLD c and OLD b (both 10) -> c = 10
```



States & State Diagram

- A machine having one or more storage element (latch, flip-flop etc.) can exist in multiple states, which may or may not change the functioning of the machine.
- The state is defined as the current values across all the storage element at the particular instance
- The state diagram helps us understand how the machine changes its state based on the input and what kind of output it generates.

State Diagram





Sequential Circuits

- Sequential circuits are those circuits whose output depends not only on the input but also on the states that exist in the storage elements.
- Combinatorial circuits only use the input and each input is independent of each other, while in sequential circuits, the output depends on the input as well as the state the machine is in.
- A Counter is an example of a sequential circuit.



Tut Question

Construct a machine which is going to transition between 4 directions North, South, East and West with the input being to turn either left or right.